



2026 Tanner Crab CIE Review Management and Tier Considerations

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NOAA FISHERIES

Crab Management Tiers 1-4

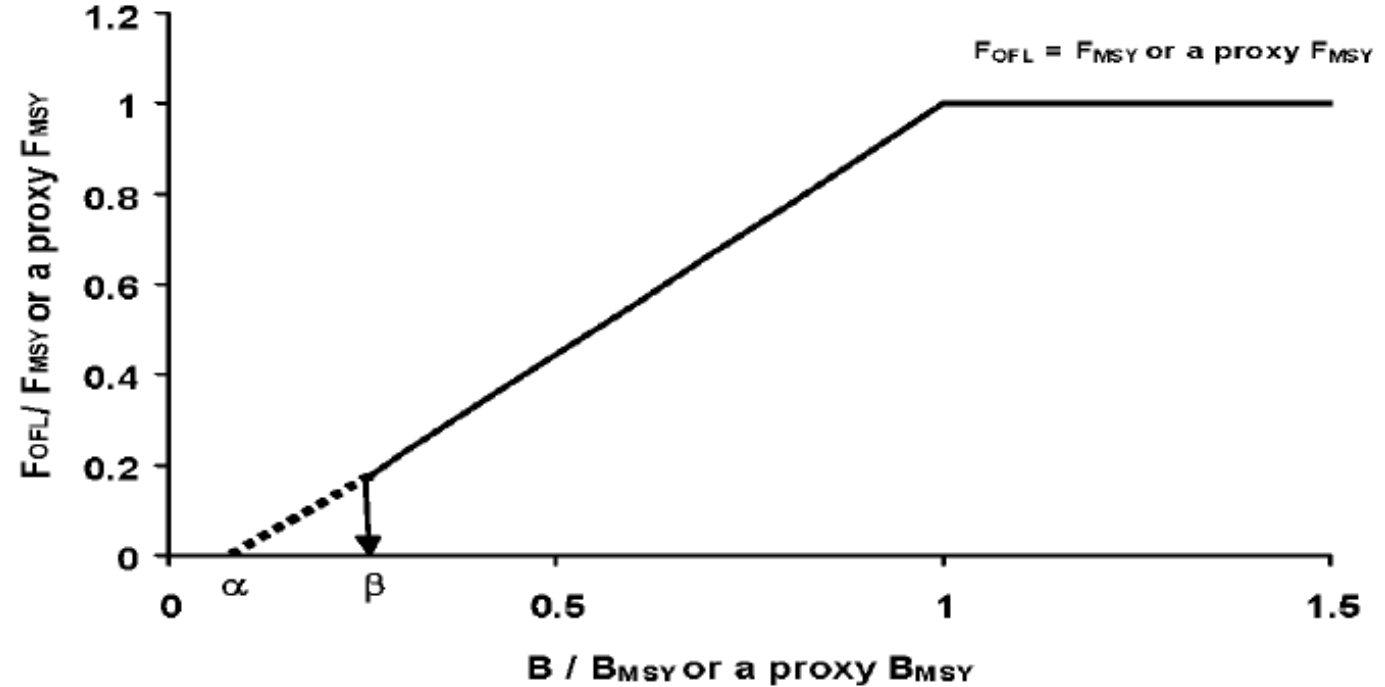
Tier 3

- no reliable S-R relationship but
- can estimate proxies for F_{MSY} and B_{MSY}
- proxies are SPR-related quantities
 - $B_{MSY} = B_{35\%} = 0.35 \cdot B_{100} = 0.35 \cdot \bar{R} \cdot SPR_{100}$
 - $F_{MSY} = F_{35\%}$
- originally developed for groundfish (Clark, 1983)
- the OFL accounts for all losses not attributable to M
- $maxABC$ determined by $p^* = p(OFL > OFL_{TRUE}) = 0.49$
- ABC typically determined by applying a buffer to OFL

Tier 4

- life-history and recruitment info do not rise to Tier 3
- but reliable estimates of current survey biomass and M do
- proxies are:
 - $F_{MSY} = \gamma \cdot M$
 - $B_{MSY} = \bar{B}$

F_{OFL} - Control Rule

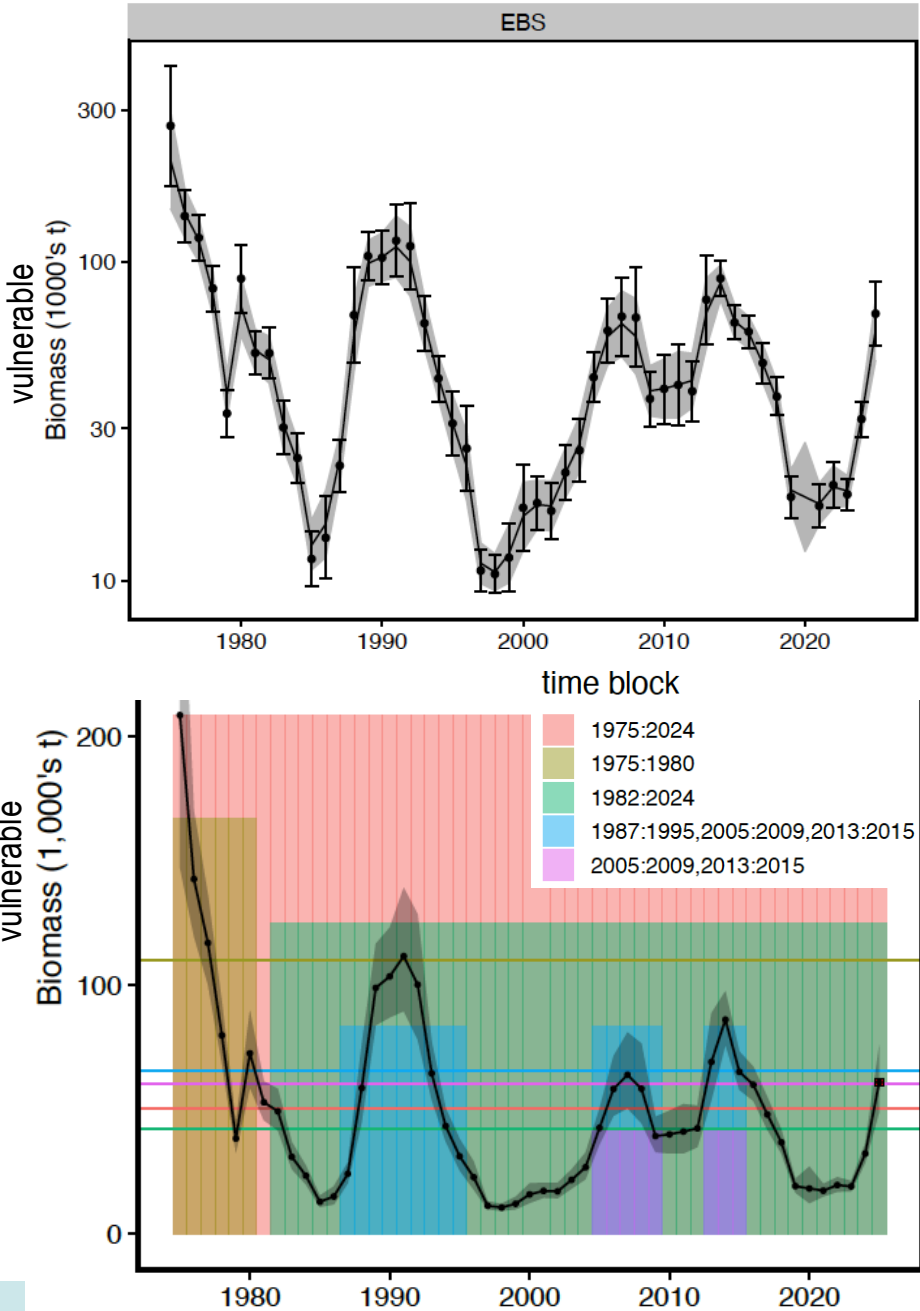


Tier 4 Backup Model

- $M = 0.23 \text{ yr}^{-1}$
- B and $\bar{B}$ based on rema-smoothed “vulnerable” biomass time series
- Potential averaging times for “vulnerable” biomass
 - 1982-2024
 - 1975-1980
 - 1975-2024
 - 1987-95, 2006-09, 2013-15
 - 2005-09, 2013-15

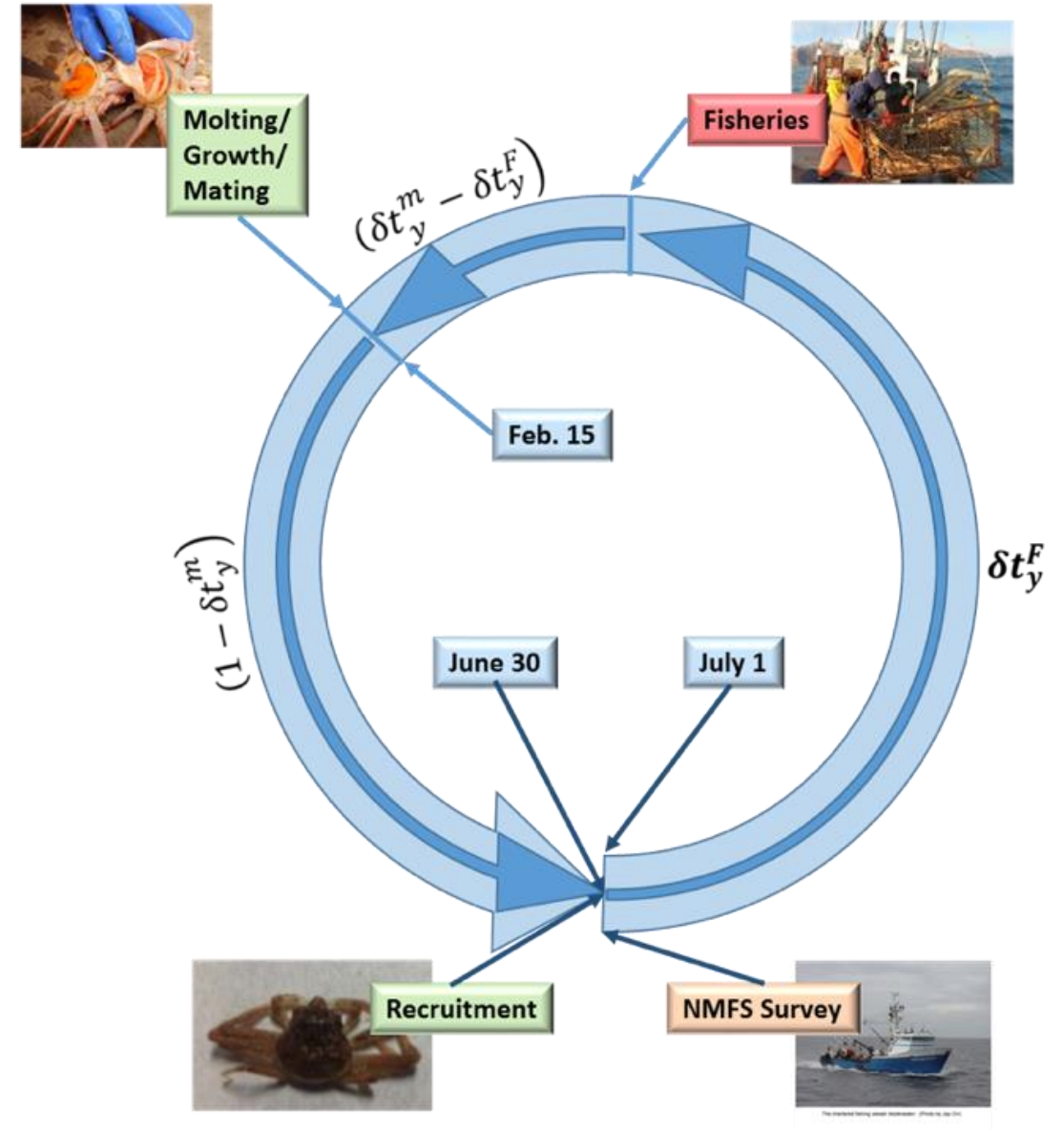
time block	M	B	B_{MSY}	status	F_{OFL}	OFL
1975:2025	0.23	61.04	50.50	1.21	0.23	12.54
1975:1980	0.23	61.04	109.90	0.56	0.12	6.71
1982:2025	0.23	61.04	42.34	1.44	0.23	12.54
1987:1995,2005:2009,2013:2015	0.23	61.04	65.88	0.93	0.21	11.62
2005:2009,2013:2015	0.23	61.04	60.42	1.01	0.23	12.54

biomass units: 1000's t



Tier 3 Assessment Model

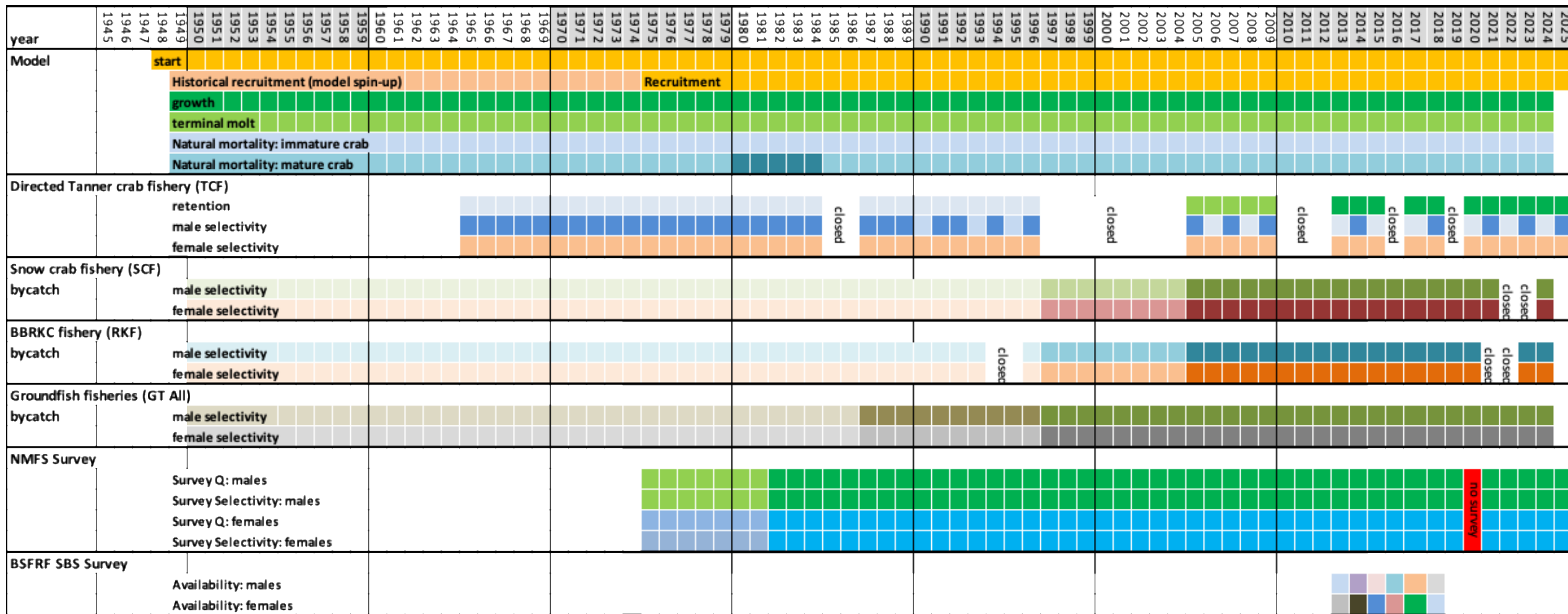
- Size-structured, two-sex model with terminal molt
 - Survey data
 - NMFS EBS shelf survey: 1975-present
 - BSFRF 2013-2018 side-by-side haul studies
 - growth data
 - male maturity data
 - Fishery data
 - directed fishery (areas combined)
 - retained catch
 - total catch
 - bycatch in
 - snow crab fishery
 - BBRKC fishery
 - groundfish fisheries
 - Estimates:
 - Annual recruitment
 - Annual numbers-at-size (M,F)
 - mature biomass (MMB, MFB)



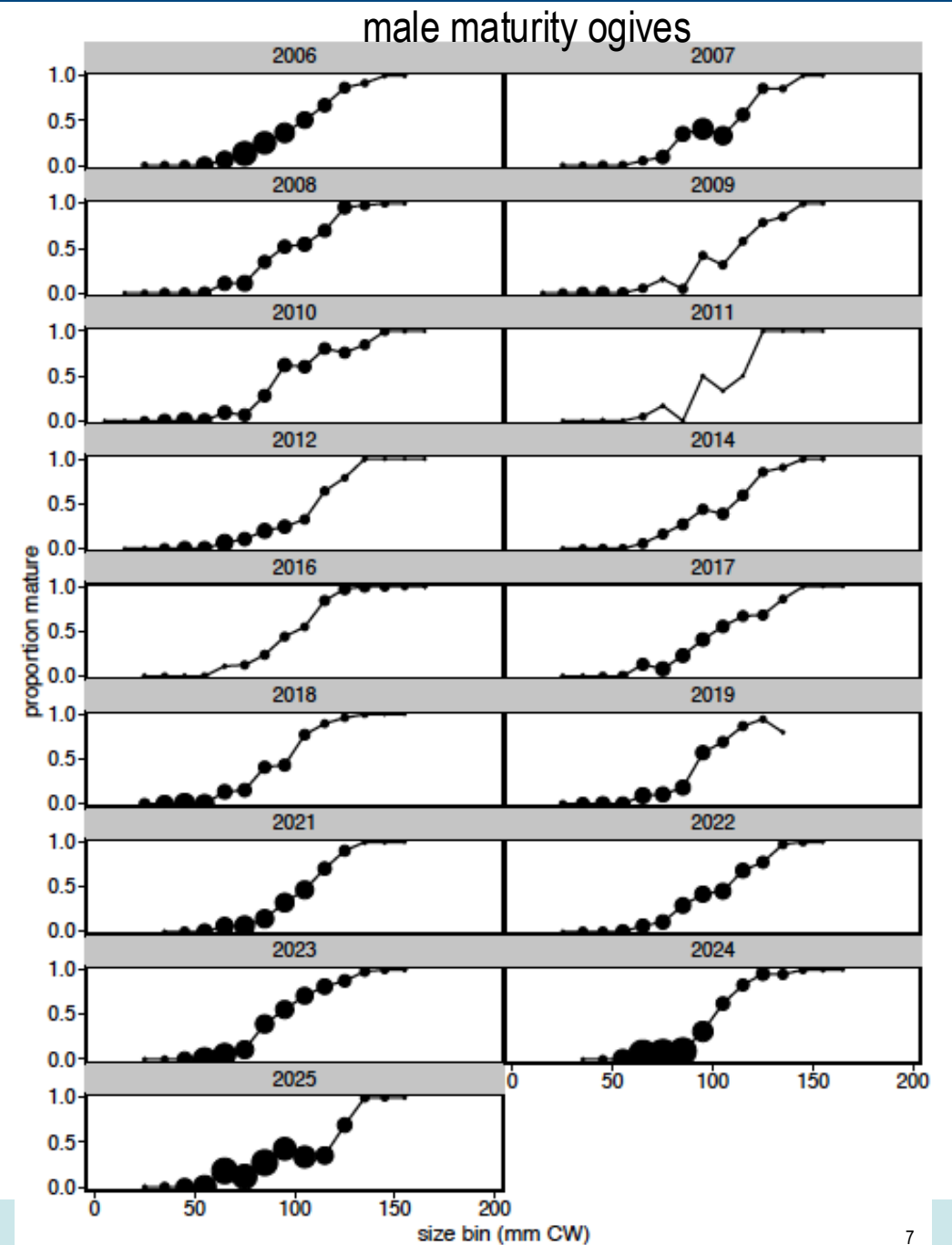
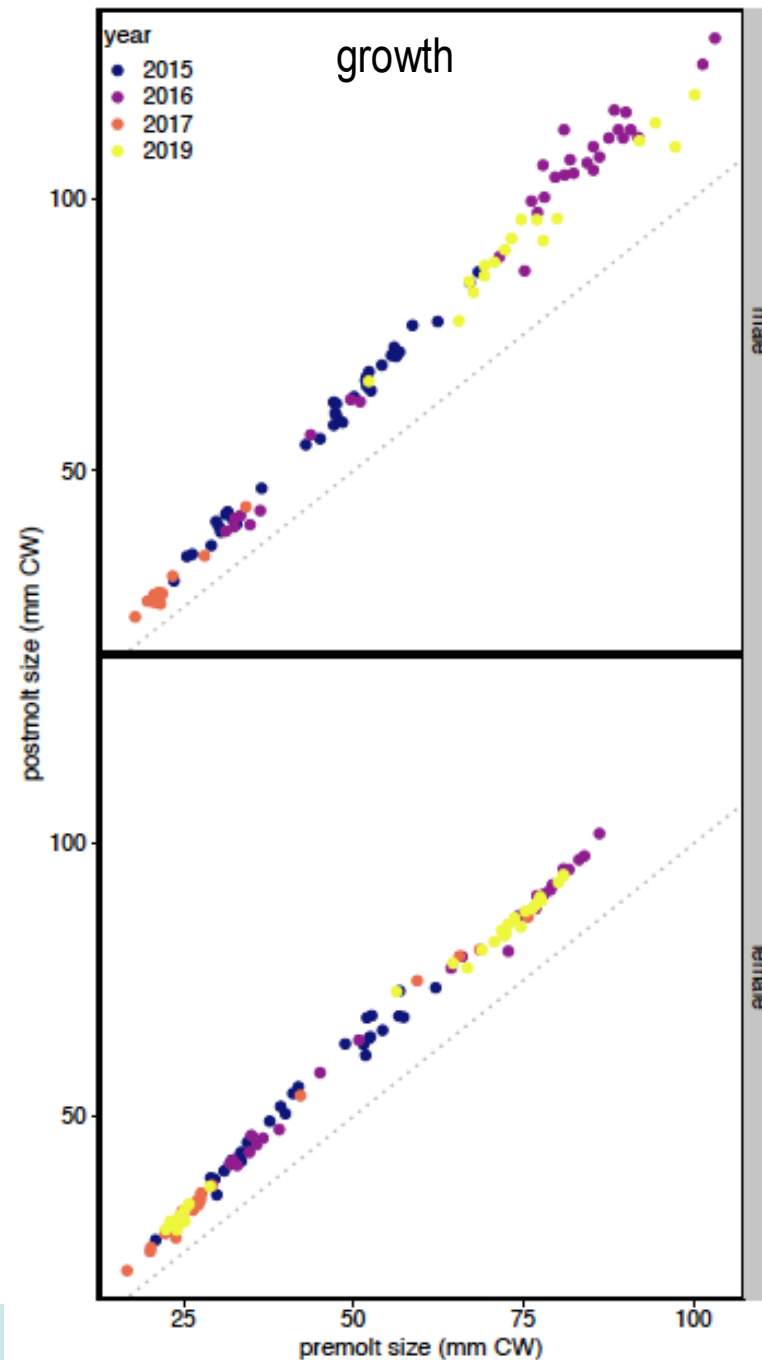
Assessment time frames: data

[illegible]

Assessment time frames: model processes

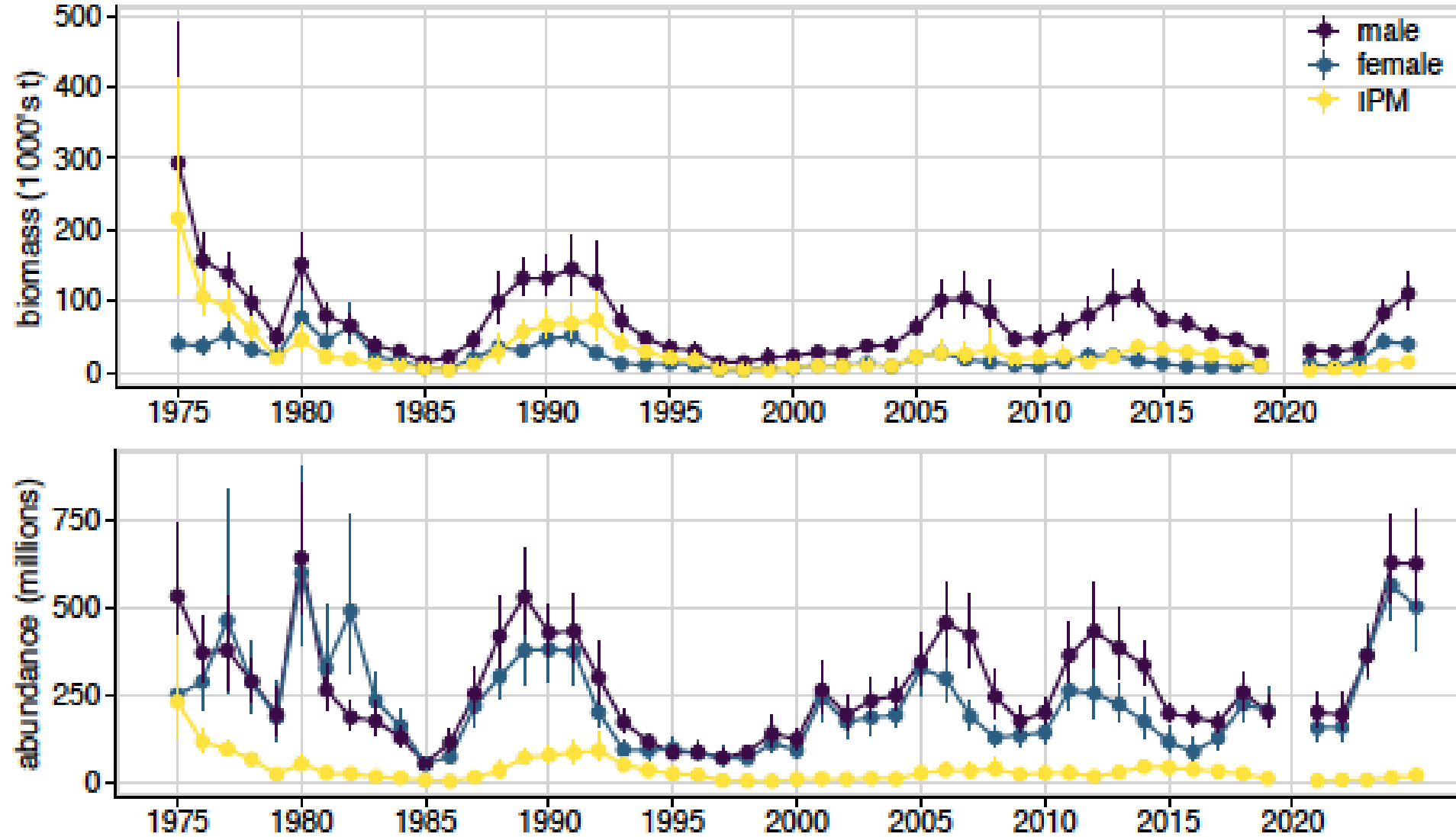


Life History Data



NMFS EBS Survey Data

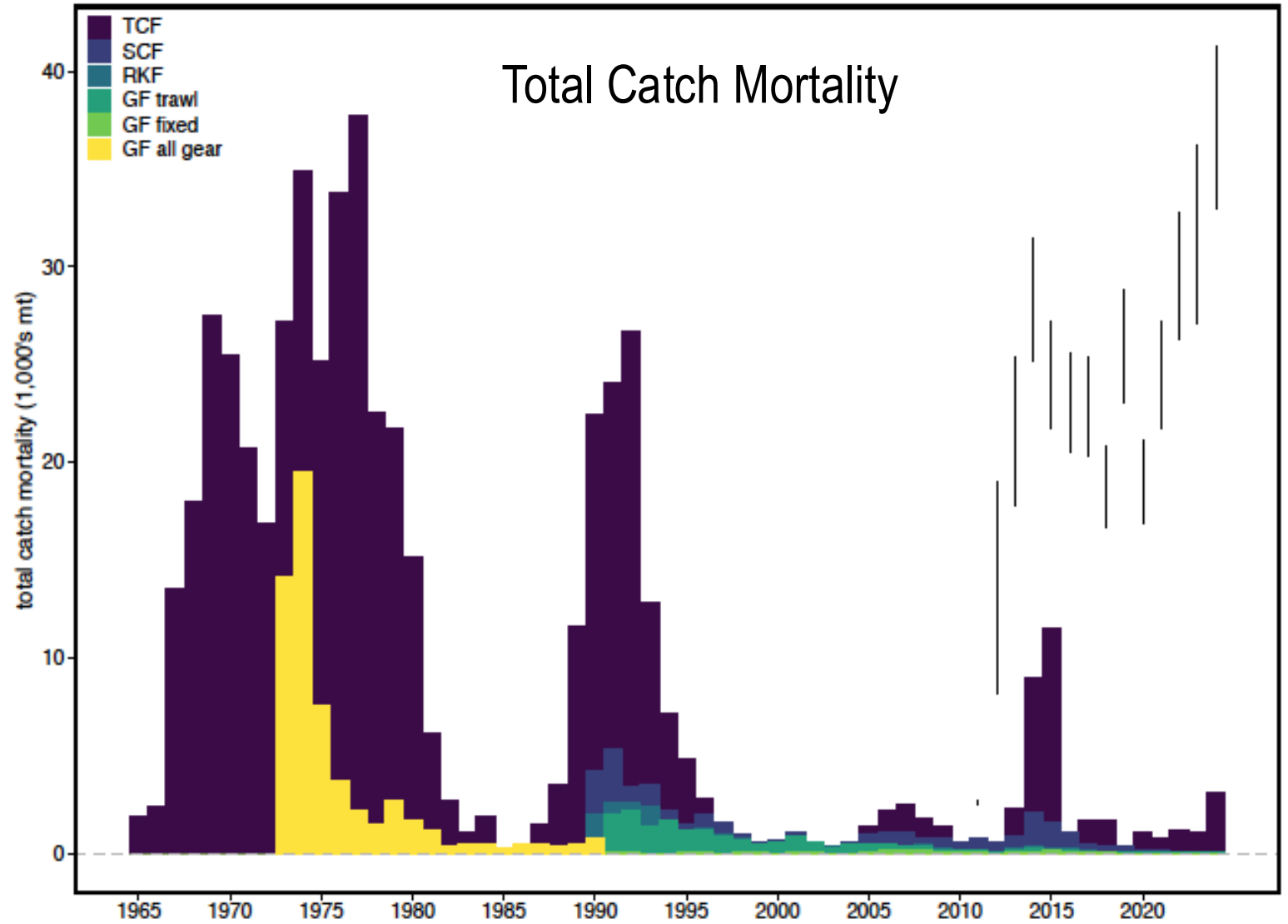
IPM: Industry-preferred males (> 125 mm, 5 inch CW)



OFLs and ABCs

vertical lines:

- upper end: federal OFL
- lower end: federal ABC



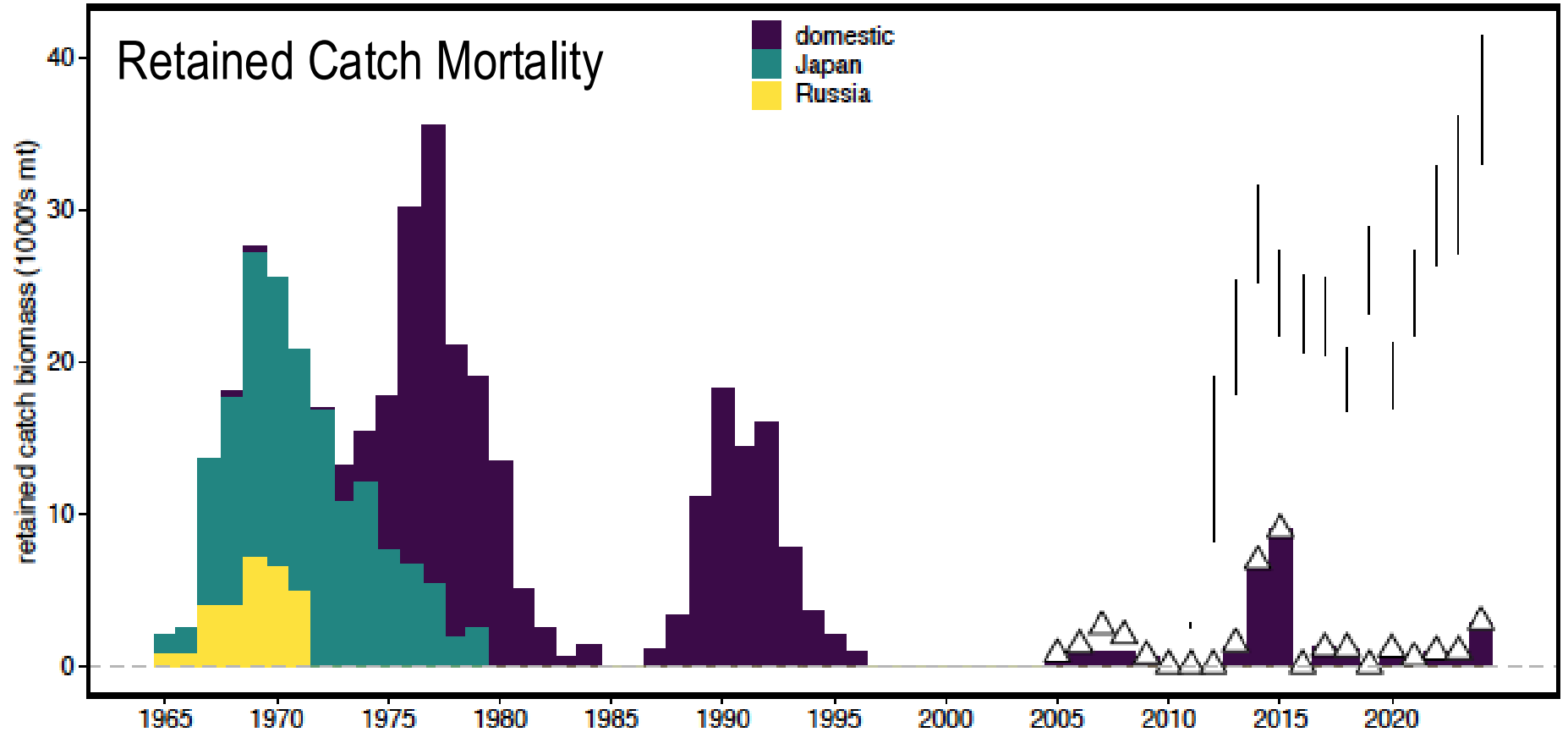
OFLs, ABCs, and TACs

vertical lines:

- upper end: federal OFL
- lower end: federal ABC

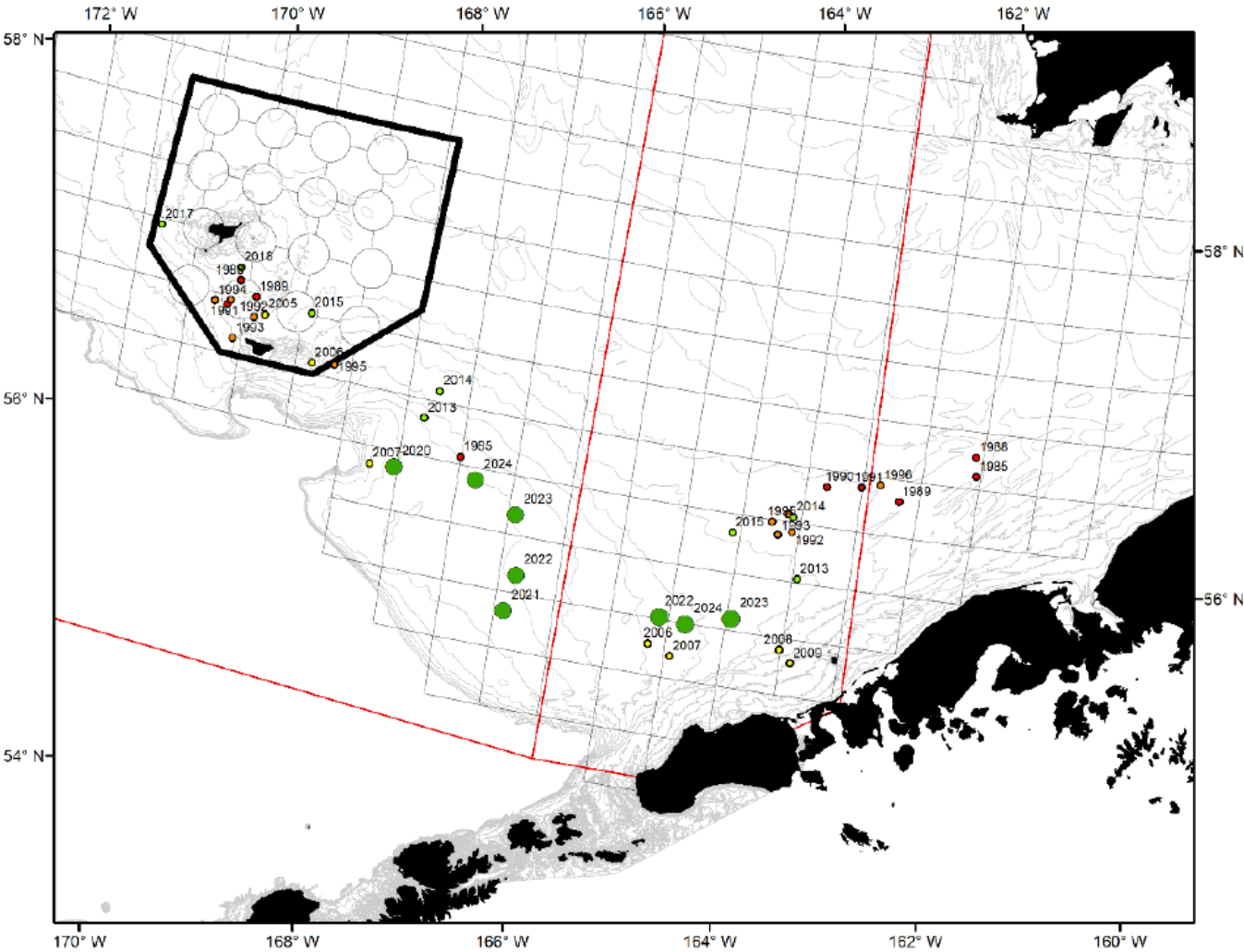
triangles:

- combined State TAC



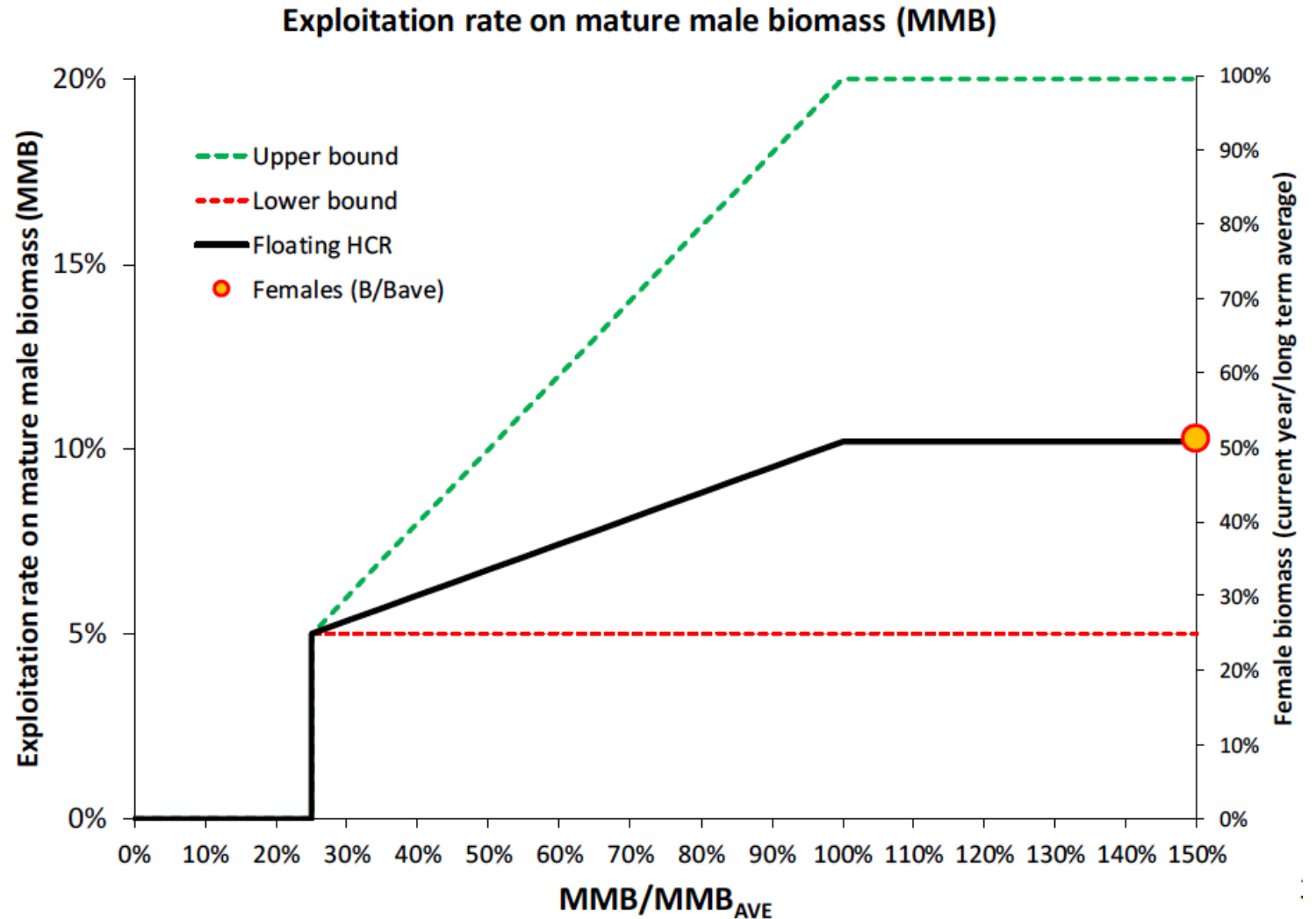
Retained catch and effort

slide 8 from TAC setting

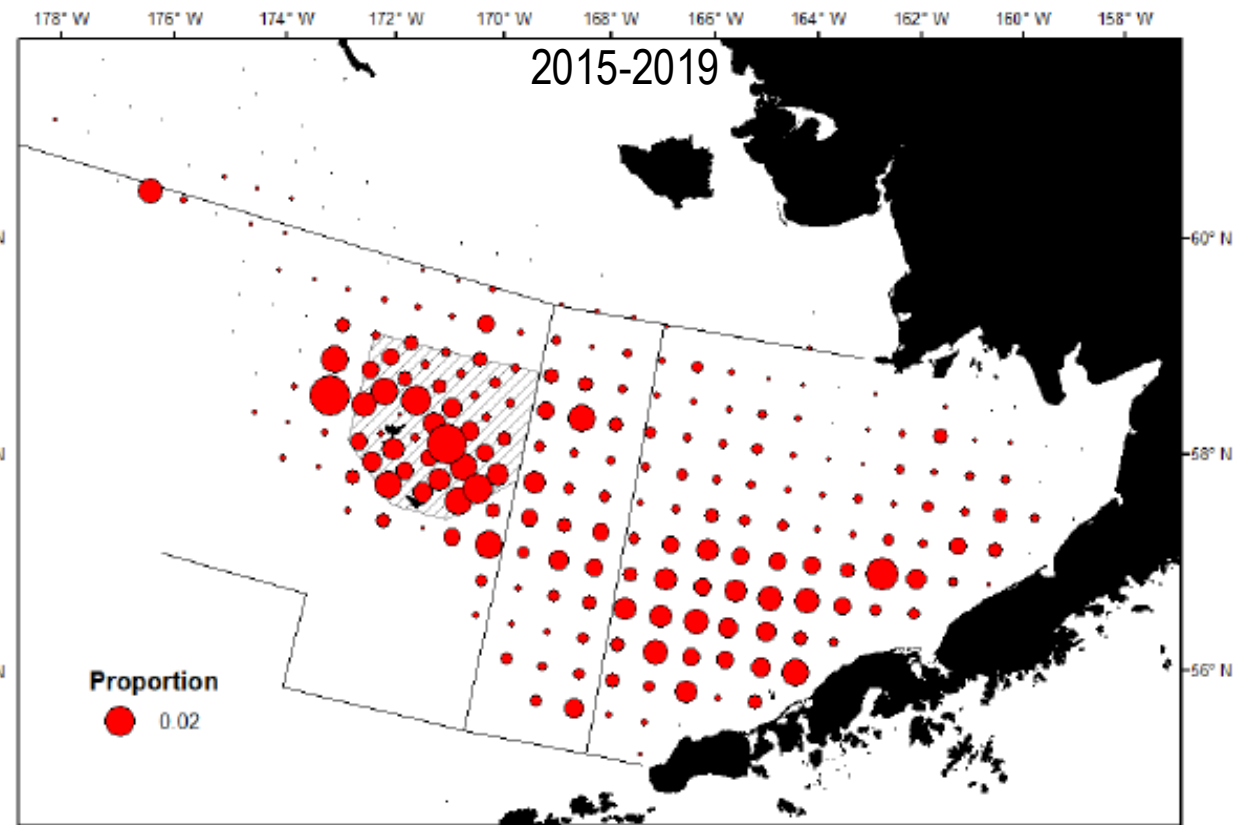
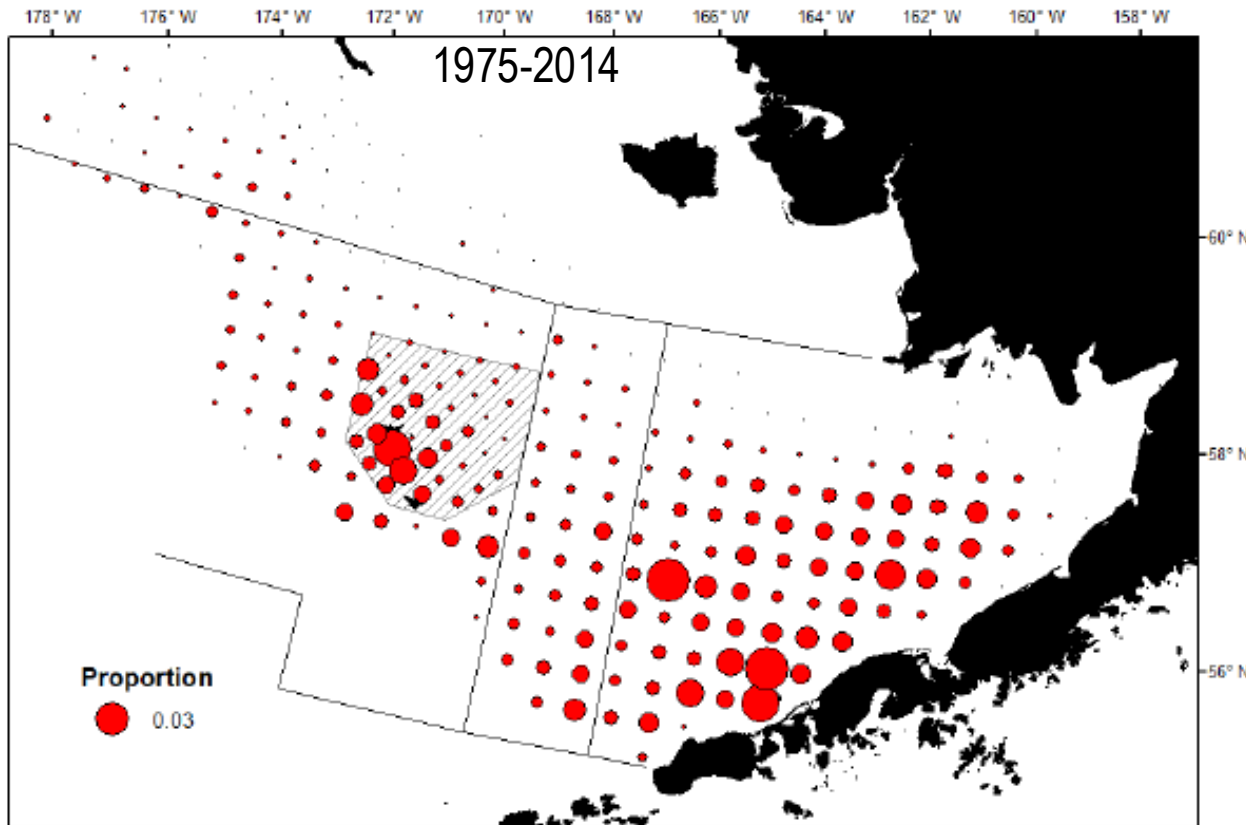


State Harvest Strategy

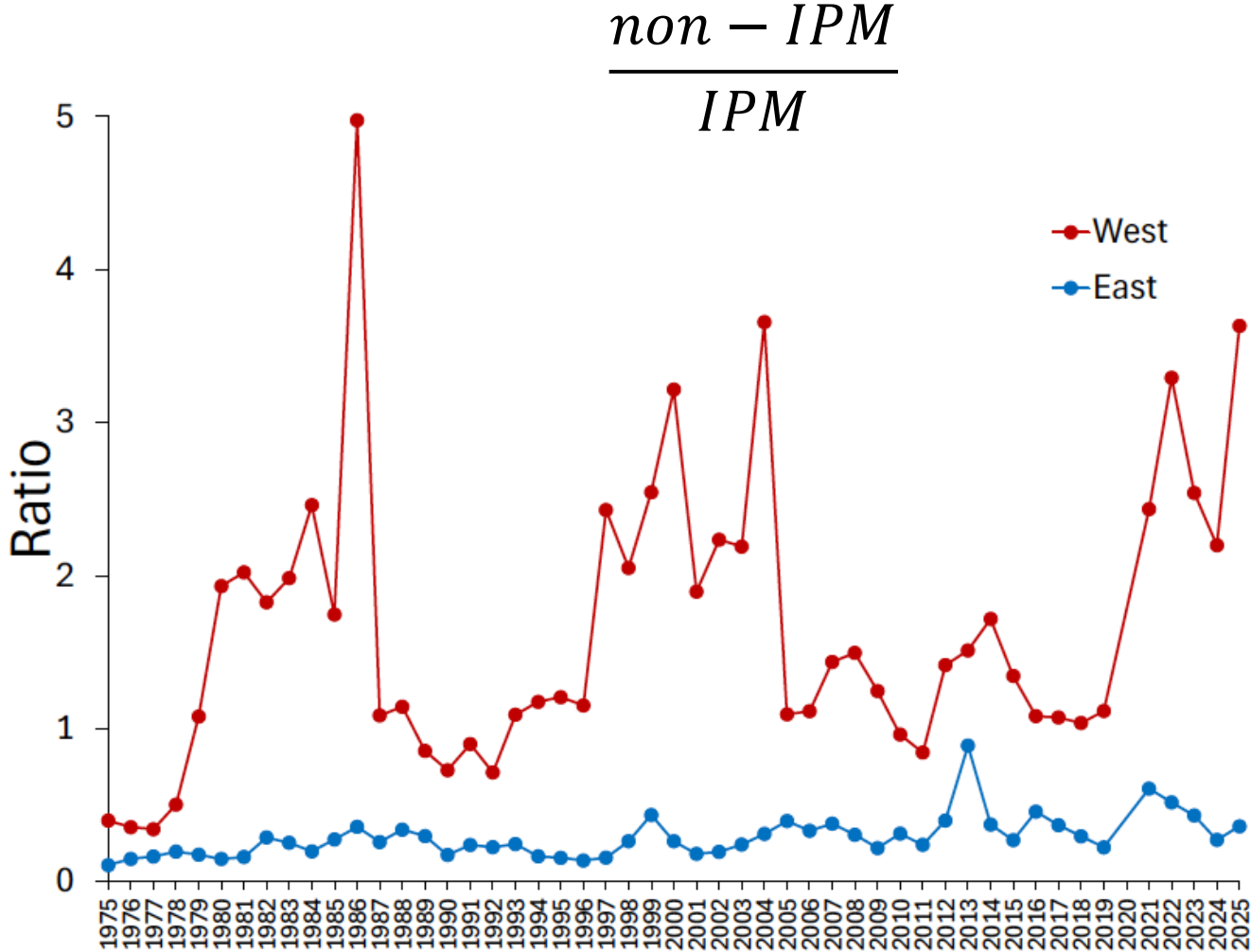
- Exploitation rates on MMB
 - 5% lower bound
 - 20% upper bound
 - Applied by State area (E/W 166W)
- “Floating” control rule determined by female biomass (entire EBS)
- Additional considerations: caps on exploitation rate for “exploitable” legal males
 - 100% new shell crab > 5in CW
 - 40% old shell crab > 5 in CW
 - cap: 50% by weight
- Requires:
 - MFB for entire EBS
 - MMB by E/W 166W
 - 5” male abundance by shell condition E/W 166W
 - Avg weight of 5” males E/W 166W



East/West Differences: MMB



East/West Differences



Courtesy of Ben Daly, ADFG